

ISAAC (ZAC) ADJEI

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PROFILE

Engineering student at Aston University (Predicted First Class) with strong foundations in data engineering, machine learning and analytics. Builds data pipelines, implements ML models and creates insightful visualisations. Seeking data engineering and AI/ML internships.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, C, C++, Java
Frontend: React, Next.js, Tailwind CSS, HTML5, CSS3, Framer Motion
Backend: Node.js, FastAPI, PostgreSQL, Redis, REST APIs, JWT Auth, RBAC
Cloud Devops: AWS, Docker, Vercel, Cloudflare, GitHub Actions, CI/CD, Terraform
Tools: Git, GitHub, VS Code, Postman, Figma

PROJECTS

PHAEMOS: Smart Maintenance Platform 2026, ongoing

Next.js 14, FastAPI, PostgreSQL, Redis, Docker, ESP32, Arduino, STM32, scikit-learn

- End-to-end IoT predictive maintenance platform with ESP32, Arduino and STM32 nodes
- FastAPI backend processing 1000+ telemetry readings/day with sub-200ms response times
- Isolation Forest ML anomaly detection triggering automated maintenance tickets

[Website ↗](#) | [GitHub ↗](#)

Embedded Sensor Networks 2025

STM32, Arduino, C++, I2C, SPI, MQTT, InfluxDB

- Multi-node sensor network collecting environmental data (temp, humidity, pressure)
- STM32 master coordinating 8+ slave nodes via I2C with error handling
- Real-time data streaming via MQTT to InfluxDB time-series database

Quantitative Trading Simulator 2025

Python, Pandas, NumPy, Backtrader, Yahoo Finance API

- Backtesting engine for momentum and mean-reversion strategies
- Implemented risk management with stop-losses and position sizing
- Portfolio analytics with Sharpe ratio, drawdown and volatility calculations

EDUCATION

Aston University, Birmingham Oct 2024 - Jul 2028

BEng (Hons), Electronic Engineering and Computer Science · **Predicted: First Class**

Top 40 Finalist - Black Heritage Undergraduate of the Year Award 2026

Relevant Modules: Internet Applications and Databases, Foundations of AI and Data Science, Introductory Programming for Digital Science (Python), Digital Circuit Design, Object-Oriented Programming (C++)

EXPERIENCE

TargetJobs / Sky 2026

Top 40 Finalist - Black Heritage Undergraduate of the Year

- Recognised among thousands of applicants for academic excellence and leadership potential
- Assessment centre experience including case studies and technical interviews

Cancer Research UK 2022-2024

Shop Volunteer

- Customer service and retail operations in charity shop environment
- Cash handling, stock management and visual merchandising